Machine Learning Applications

Text Processing

Introduction

Text Processing

- Mathematical models work with numbers, not text
- Text data need to be converted to numbers before it can be fed into mathematical models
- Convert text to numbers via Text Featurization

Terminologies

Document

- A collection of words
- Article, Email, SMS etc

Corpus

- A collection of documents of a certain theme (e.g. movie reviews, product reviews)
- A machine learning model for textual data usually depends on a corpus to learn the nuances of a subject matter

Terminologies

Which is a document?

A) “Hi, can we meet at 3pm?” ✔

B) A full news article about climate change ✔

C) An email with subject and body ✔

D) A single word: “Hello” ✔

E) A paragraph from a book ✔

G) A photo (no text) ✘

H) A tweet / social media post ✔

Terminologies

Which is a corpus?

A) A folder containing 10,000 movie reviews ✔

B) A single email ✘

C) A dataset of customer feedback messages collected over 1 year ✔

D) A WhatsApp message ✘

E) A collection of news articles about AI ✔

F) A single paragraph from a book ✘

G) A database of product reviews from an e-commerce site ✔

Terminologies

Term / Token

- Smallest processable unit (e.g. a word within a document)

Vocabulary

- The number of unique terms in a corpus
- The Vocabulary Size is the number of features that our model needs to learn from

Terminologies

Corpus

"I like AI"

"I like machine learning"

"AI is powerful"

What's the Vocabulary Size?

7

Corpus:

"I like AI."

"She likes AI!"

What's the Vocabulary Size?

Is "like" and "likes" same?

Text Preparation

Text Cleansing

- Remove punctuations (e.g. full-stops, commas, exclamations)
- Convert all words to lowercase or uppercase (for consistent comparisons)
- Remove formatting (e.g. HTML tags)

Tokenization

- Tokenization splits a document into tokens or terms
- NLTK, a Natural Language Processing (NLP) tool, can perform tokenization on a string as shown

```
from nltk import word_tokenize

text = 'Hello this is a test.'

print(word_tokenize(text))
```

```
['Hello', 'this', 'is', 'a', 'test', '.']
```

Terms

Stemming and Lemmatization

Both Stemming and Lemmatization shorten a word to its root form

Stemming

- Uses rule-based heuristics
- Cuts off prefixes and/or ends of words
- Quicker, but the shorten word might not make sense

Lemmatization

- Uses a vocabulary for its transformation
- Considers the context and return an actual word in the vocabulary

Stemming example

```
from nltk import word_tokenize
from nltk.stem import SnowballStemmer

text = 'he likes cats and dogs, and teaching machines to learn'

stem = SnowballStemmer(language='english')

print([stem.stem(token) for token in word_tokenize(text)])
```

['he', 'like', 'cat', 'and', 'dog', ',', 'and', 'teach', 'machin', 'to', 'learn']

Not a word in the dictionary

Lemmatization example

```
from nltk import word_tokenize
from nltk.stem import WordNetLemmatizer

text = 'he likes cats and dogs, and teaching machines to learn'

lm = WordNetLemmatizer()

print([lm.lemmatize(token, 'v') for token in word_tokenize(text)])
```

['he', 'like', 'cat', 'and', 'dog', ',', 'and', 'teach', 'machine', 'to', 'learn']

An actual word in the dictionary

Examples

	likes	studies	better	running
Stemming	like	studi	better	run
Lemmatization	like	study	good	run

Stop Words

- Stop Words are words that are very common and deemed as not providing differentiating value when fed into models

- In Text Processing, it is a common practice to remove stop words from our corpus before doing further processing

- However, there is no universal list of stop words; every Natural Language Processing (NLP) tool has its own

Stop Words

- NLTK's English stop words are shown below

```
from nltk.corpus import stopwords

print(stopwords.words('english'))
```

```
['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've", "you'll", "you'd", 'your', 'yours',
'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', "she's", 'her', 'hers', 'herself', 'it', "it's", 'its', 'itself', 'they',
'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', "that'll", 'these', 'those', 'am', 'is',
'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'a', 'an', 'the',
'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into',
'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under',
'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few',
'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can',
'will', 'just', 'don', "don't", 'should', "should've", 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', "aren't", 'couldn',
"couldn't", 'didn', "didn't", 'doesn', "doesn't", 'hadn', "hadn't", 'hasn', "hasn't", 'haven', "haven't", 'isn', "isn't",
'ma', 'mightn', "mightn't", 'mustn', "mustn't", 'needn', "needn't", 'shan', "shan't", 'shouldn', "shouldn't", 'wasn',
"wasn't", 'weren', "weren't", 'won', "won't", 'wouldn', "wouldn't"]
```

Stop Words

- An example where stop words and punctuations are filtered away using NLTK's list of stop words

```
from nltk import word_tokenize
from nltk.corpus import stopwords
import string

text = 'he likes cats and dogs, and teaching machines to learn'

stops = stopwords.words('english')

punc = str.maketrans('', '', string.punctuation)
text_no_punc = text.translate(punc)

terms = [token for token in word_tokenize(text_no_punc) if token not in stops]

print(terms)
```

['likes', 'cats', 'dogs', 'teaching', 'machines', 'learn']

Information Extraction

Part-of-Speech & Named Entities

Part-of-Speech

- A sentence can have extra grammatical properties tagged to it. Those tagged properties are called Part-of-Speech (POS) tags
- The POS tags of each word in our sentence

```python
import nltk
from nltk.tokenize import word_tokenize

text = word_tokenize("Mary has a little lamb")
pos = nltk.pos_tag(text)
print(pos)
```

[('Mary', 'NNP'), ('has', 'VBZ'), ('a', 'DT'), ('little', 'JJ'), ('lamb', 'NN')]

Part–of-Speech (POS)

Part-of-Speech

- Meanings of the POS tags in our sentence

Word	Abbreviation	Meaning
Mary	NNP	Proper noun, Singular (e.g. Sarah)
has	VBZ	Verb, Present Tense
a	DT	Determiner (e.g. a, the)
little	JJ	Adjective (e.g. large, red)
lamb	NN	Noun, Singular (e.g. cat, tree)

✤ A complete table of abbreviation/meaning of NLTK's POS tags can be found here - https://www.guru99.com/pos-tagging-chunking-nltk.html

Named Entity

- Named Entities, such as dates, company names, locations can further be identified within a sentence
- Such add-on data provides your application, e.g. chatbot, with useful context to better understand a sentence

To further elaborate on the geographical trends, North America [LOC] has procured more than 50% [PERCENT] of the global share in 2017 [DATE] and has been leading the regional landscape of AI [GPE] in the retail market. The U.S. [GPE] has a significant credit in the regional trends with over 65% [PERCENT] of investments (including M&As, private equity, and venture capital) in artificial intelligence technology. Additionally, the region is a huge hub for startups in tandem with the presence of tech titans, such as Google [ORG], IBM [ORG], and Microsoft [ORG].

Text Featurization

Bag of Words (BOW)

Text Featurization

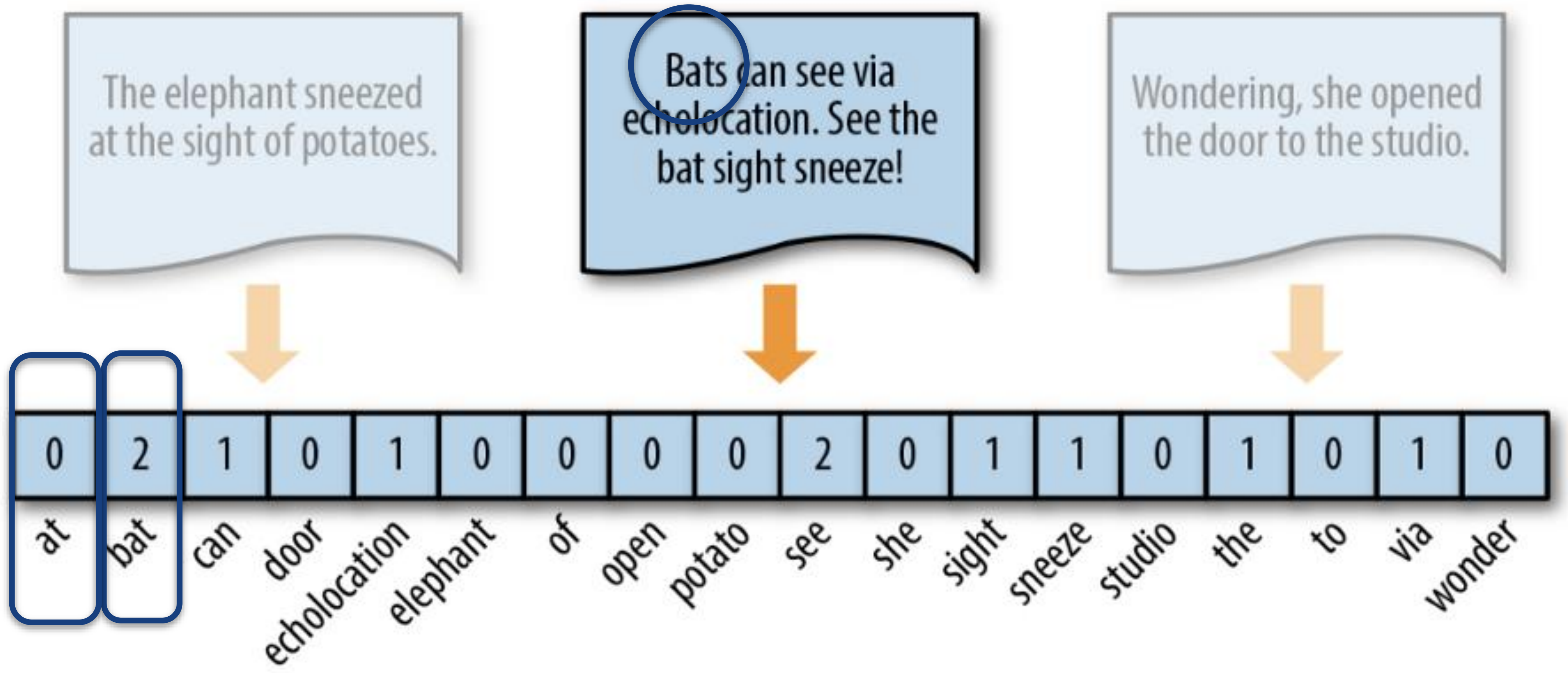
- Mathematical Models only works with numbers
- In order for text to be useful to our models, those text need to be converted to some meaningful numbers (features)
- Common techniques in generating features for text
  - Bag of Words (BOW)
  - TF-IDF

Bag of Words (BOW)

- Bag of Words counts the frequencies of words in a corpus
- BOW discards grammar such as present and past tenses, hence the context might be lost
- BOW also discards word-order, hence understanding the meaning of a sentence is difficult
  - “He genuinely needs to do that.”
  - “He needs to do that genuinely.”

Bag of Words (BOW)

- Each document has a vector to track words found within itself
- Each unique word in the Corpus is mapped to the same position in all the vectors



BOW example

doc1: John has some cats
doc2: Cats, being cats, eat fish
doc3: I ate a big fish

stop-words removal

[has, some, being, I, a]

doc1: John cats
doc2: Cats cats eat fish
doc3: ate big fish

lemmatization

doc1: john cat
doc2: cat cat eat fish
doc3: eat big fish

BOW example

doc1: john cat
doc2: cat cat eat fish
doc3: eat big fish

Vocabulary

[big, cat, eat, fish, john]

Pos-to-Word mapping for Feature Vectors

Index	0	1	2	3	4
Word	big	cat	eat	fish	john

BOW example

doc1: john cat
doc2: cat cat eat fish
doc3: eat big fish

Compute Bag-of-Words (BOW)

	big	cat	eat	fish	john
doc1	0	1	0	0	1
doc2	0	2	1	1	0
doc3	1	0	1	1	0

Feature Vectors

BOW (in code)

- Download required NLTK data – its list of stop-words and wordnet (database of English words)

- Setup Lemmatizer to shorten words to root form

```
import string
import pandas as pd
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import CountVectorizer

# can be removed if resources reside in your system
nltk.download('stopwords')
nltk.download('wordnet')

lemmatizer = WordNetLemmatizer()
stop_words = stopwords.words('english')

docs = [
    'John has some cats.',
    'Cats, being cats, eat fish.',
    'I ate a big fish.'
]
```

BOW (in code)

- Next, we filter out punctuations, stop-words, and perform Lemmatization
- str.maketrans(…) creates a mapping of punctuation-symbols to None, and doc.translate(…) performs substitution using that mapping

```
docs_clean = []
punc = str.maketrans('', '', string.punctuation)

# pre-process each document
for doc in docs:
    doc_no_punc = doc.translate(punc)
    words = doc_no_punc.lower().split()

    words = [lemmatizer.lemmatize(word, 'v')
                for word in words if word not in stop_words]

    docs_clean.append(' '.join(words))
```

BOW (in code)

- Finally, we generate feature vectors for all documents and align them with the vocabulary in our corpus

```
# generate feature vectors using BOW
bow = CountVectorizer()

feature_vectors = bow.fit_transform(docs_clean).toarray()
vocab = bow.get_feature_names()

df = pd.DataFrame(data=feature_vectors,
                  index=['doc1', 'doc2', 'doc3'],
                  columns=vocab)

print(df)
```

	big	cat	eat	fish	john
doc1	0	1	0	0	1
doc2	0	2	1	1	0
doc3	1	0	1	1	0

Entire code

```
import string
import pandas as pd
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import CountVectorizer

# can be removed if resources reside in your system
nltk.download('stopwords')
nltk.download('wordnet')

lemmatizer = WordNetLemmatizer()
stop_words = stopwords.words('english')

docs = [
    'John has some cats.',
    'Cats, being cats, eat fish.',
    'I ate a big fish.'
]

# data cleansing
docs_clean = []
punc = str.maketrans('', '', string.punctuation)
for doc in docs:
    doc_no_punc = doc.translate(punc)
    words = doc_no_punc.lower().split()

    words = [lemmatizer.lemmatize(word, 'v')
             for word in words if word not in stop_words]

    docs_clean.append(' '.join(words))
```

→

```
# generate feature vectors using BOW
bow = CountVectorizer()

feature_vectors = bow.fit_transform(docs_clean).toarray()
vocab = bow.get_feature_names()

df = pd.DataFrame(data=feature_vectors,
                  index=['doc1', 'doc2', 'doc3'],
                  columns=vocab)

print(df)
```

Mini Exercise - 1

Bag of Words

Text Featurization

TF-IDF

TF-IDF

- TF-IDF measures the importance of terms with respect to documents within a corpus

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t, c)$$

How frequent a term appears in a document

How infrequent the same term appears in our corpus

TF-IDF

- Term Frequency (TF): The number of times a term appears in a document
- TF can easily be implemented using Bag of Words

$$\text{TF-IDF} = \text{TF}(t, d) \times \text{IDF}(t, c)$$

No. of times a term t appears in a doc d

TF-IDF

- Inverse Document Frequency (IDF): The IDF of a word is the inverse of the number of times it appears in the corpus at least once
- The fewer times a term appears in the corpus, the more important it is considered

$$\text{TF-IDF} = \text{TF}(t, d) \times \text{IDF}(t, c)$$

The inverse of the no. of times term t appears in corpus c at least once

TF-IDF

- There is only one IDF value for each unique term in a corpus
- Compute IDF with this formula

term

corpus

total no. of documents
in our corpus

$$IDF(t, c) = \ln \frac{1 + n}{1 + df(t, c)} + 1$$

natural log

document frequency (no.
of times the term appears
in our corpus at least once

Normalized TF-IDF

- A long document may have high frequencies for certain terms due to its length
- Given a query string, a long document may seem more relevant than a short document if compared using absolute term frequencies
- Normalization gives us relative term frequencies instead of absolute term counts

Normalized TF-IDF

- SKLearn adds normalization as an extra step to our earlier TF-IDF formula by dividing each TF-IDF vector with its Euclidean Norms

TF-IDF vector of a document

$$v_{norm} = \frac{v}{\| v \|^2} = \frac{v}{\sqrt{(v_1)^2 + (v_2)^2 + \ldots + (v_n)^2}}$$

Euclidean distance of the TF-IDF vector

TF-IDF example

doc1: john cat
doc2: cat cat eat fish
doc3: eat big fish

Vocabulary → [big, cat, eat, fish, john]

Term Frequency (TF) = our BOW feature vectors

	big	cat	eat	fish	john
doc1	0	1	0	0	1
doc2	0	2	1	1	0
doc3	1	0	1	1	0

TF-IDF example

- To normalize our Term Frequency (TF) vectors, divide each count with the total number of counts for each document

	big	cat	eat	fish	john	Counts
doc1	0	1	0	0	1	2
doc2	0	2	1	1	0	4
doc3	1	0	1	1	0	3

TF

Normalization

	big	cat	eat	fish	john
doc1	0	1/2	0	0	1/2
doc2	0	2/4	1/4	1/4	0
doc3	1/3	0	1/3	1/3	0

Normalized TF

TF-IDF example

doc1: john cat
doc2: cat cat eat fish
doc3: eat big fish

Vocabulary

[big, cat, eat, fish, john]

Document Frequency (DF) for each term in our corpus

Vocab	Document Frequency (DF)
big	1
cat	2
eat	2
fish	2
john	1

TF-IDF example

- Computing IDF for our vocabulary

term

corpus

total no. of documents in corpus

$$\mathrm{IDF}(t, c) = \ln \frac{1 + n}{1 + \mathrm{df}(t, c)} + 1$$

natural log

document frequency (no. of time a term t appears in the corpus c at least once)

The IDF formula

Vocab	Document Frequency (DF)	Inverse Document Frequency (IDF)
big	1	$ln(\frac{1+3}{1+1}) + 1 =$ ln(4/2) + 1 = 1.693
cat	2	$ln(\frac{1+3}{1+2}) + 1 =$ ln(4/3) + 1 = 1.288
eat	2	ln(4/3) + 1 = 1.288
fish	2	ln(4/3) + 1 = 1.288
john	1	ln(4/2) + 1 = 1.693

Computed IDF for our corpus

TF-IDF example

- Computing TF-IDF of each word in each document

	big	cat	eat	fish	john
doc1	0	1/2	0	0	1/2
doc2	0	2/4	1/4	1/4	0
doc3	1/3	0	1/3	1/3	0

Normalized TF

×

Vocab	IDF
big	ln(4/2) + 1 = 1.693
cat	ln(4/3) + 1 = 1.287
eat	ln(4/3) + 1 = 1.287
fish	ln(4/3) + 1 = 1.287
john	ln(4/2) + 1 = 1.693

IDF

	big	cat	eat	fish	john
doc1	0	$\boxed{1/2 * 1.287}$ = 0.644	0	0	1/2 * 1.693 = 0.846
doc2	0	2/4 * 1.287 = 0.644	1/4 * 1.287 = 0.322	1/4 * 1.287 = 0.322	0
doc3	1/3 * 1.693 = 0.564	0	1/3 * 1.287 = 0.429	1/3 * 1.287 = 0.429	0

TF-IDF

TF-IDF example

- Computing Euclidean distance for our TF-IDF feature vectors

TF-IDF vector of a document

$$v_{norm} = \frac{v}{||v||^2} = \frac{v}{\boxed{\sqrt{(v_1)^2 + (v_2)^2 + \ldots + (v_n)^2}}}$$

Euclidean distance of the TF-IDF vector

Normalized TF-IDF

	big	cat	eat	fish	john	Euclidean Distance
doc1	0	0.644	0	0	0.846	1.063
doc2	0	0.644	0.322	0.322	0	0.788
doc3	0.564	0	0.429	0.429	0	0.828

$$\sqrt{(0.644)^2 + (0.322)^2 + (0.322)^2}$$

Euclidean Distances of each feature vector

TF-IDF example

	big	cat	eat	fish	john	Euclidean Distance
doc1	0	0.644	0	0	0.846	1.063
doc2	0	0.644	0.322	0.322	0	0.788
doc3	0.564	0	0.429	0.429	0	0.828

TF-IDF

	big	cat	eat	fish	john
doc1	0	0.644/1.063 = 0.606	0	0	0.846/1.063 = 0.796
doc2	0	0.644/0.788 = 0.817	0.322/0.788 = 0.409	0.322/0.788 = 0.409	0
doc3	0.564/0.828 = 0.681	0	0.429/0.828 = 0.518	0.429/0.828 = 0.518	0

Normalized TF-IDF

TF-IDF example

- Our corpus has been transformed into numerics to be fed into learning models

doc1: john cat
doc2: cat cat eat fish
doc3: eat big fish

$\xrightarrow{\text{Vocabulary}}$ [big, cat, eat, fish, john]

$\downarrow$ TF-IDF

	big	cat	eat	fish	john
doc1	0	0.606	0	0	0.796
doc2	0	0.817	0.409	0.409	0
doc3	0.681	0	0.518	0.518	0

Normalized TF-IDF

TF-IDF (in code)

- With sklearn’s TF-IDF vectorizer, we can generate each document’s TF-IDF feature vectors via fit_transform(…)

```
tfidf = TfidfVectorizer()
feature_vectors = tfidf.fit_transform(docs_clean).toarray()
```

- Pandas can then be used to visualize the feature vectors by document

```
df = pd.DataFrame(data=feature_vectors,
                  index=['doc1', 'doc2', 'doc3'],
                  columns=tfidf.get_feature_names())

print(df)
```

	big	cat	eat	fish	john
doc1	0	0.606	0	0	0.796
doc2	0	0.817	0.409	0.409	0
doc3	0.681	0	0.518	0.518	0

TF-IDF (in code)

```
import string
import pandas as pd
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import CountVectorizer

# can be removed if resources reside in your system
nltk.download('stopwords')
nltk.download('wordnet')

lemmatizer = WordNetLemmatizer()
stop_words = stopwords.words('english')

docs = [
    'John has some cats.',
    'Cats, being cats, eat fish.',
    'I ate a big fish.'
]

# data cleansing
docs_clean = []
punc = str.maketrans('', '', string.punctuation)
for doc in docs:
    doc_no_punc = doc.translate(punc)
    words = doc_no_punc.lower().split()

    words = [lemmatizer.lemmatize(word, 'v')
                for word in words if word not in stop_words]

    docs_clean.append(' '.join(words))
```

→

```
# generate feature vectors using TF-IDF
tfidf = TfidfVectorizer()
feature_vectors = tfidf.fit_transform(docs_clean).toarray()

df = pd.DataFrame(data=feature_vectors,
                  index=['doc1', 'doc2', 'doc3'],
                  columns=tfidf.get_feature_names())

print(df)
```

Mini Exercise 2

TF-IDF

Workshop 1

TF-IDF
(Document Summarization)

Cosine Similarity

Cosine Similarity

- After TF-IDF transforms our documents into vectors, Cosine Similarity can be used to compare the similarity among them
- The Cosine Similarity between two vectors (or two documents) is a measure that calculates the cosine angle between them
- Two documents are considered similar to each other if the cosine angle between their feature vectors is small

Cosine Similarity

- The Dot Product of two vectors is given as

dot product

length of vectors

$$\vec{a} \cdot \vec{b} = \| \vec{a} \| \| \vec{b} \| \cos\theta$$

vectors

angle between the two vectors

Cosine Similarity

- The Cosine Similarity of two feature vectors is simply

feature vectors

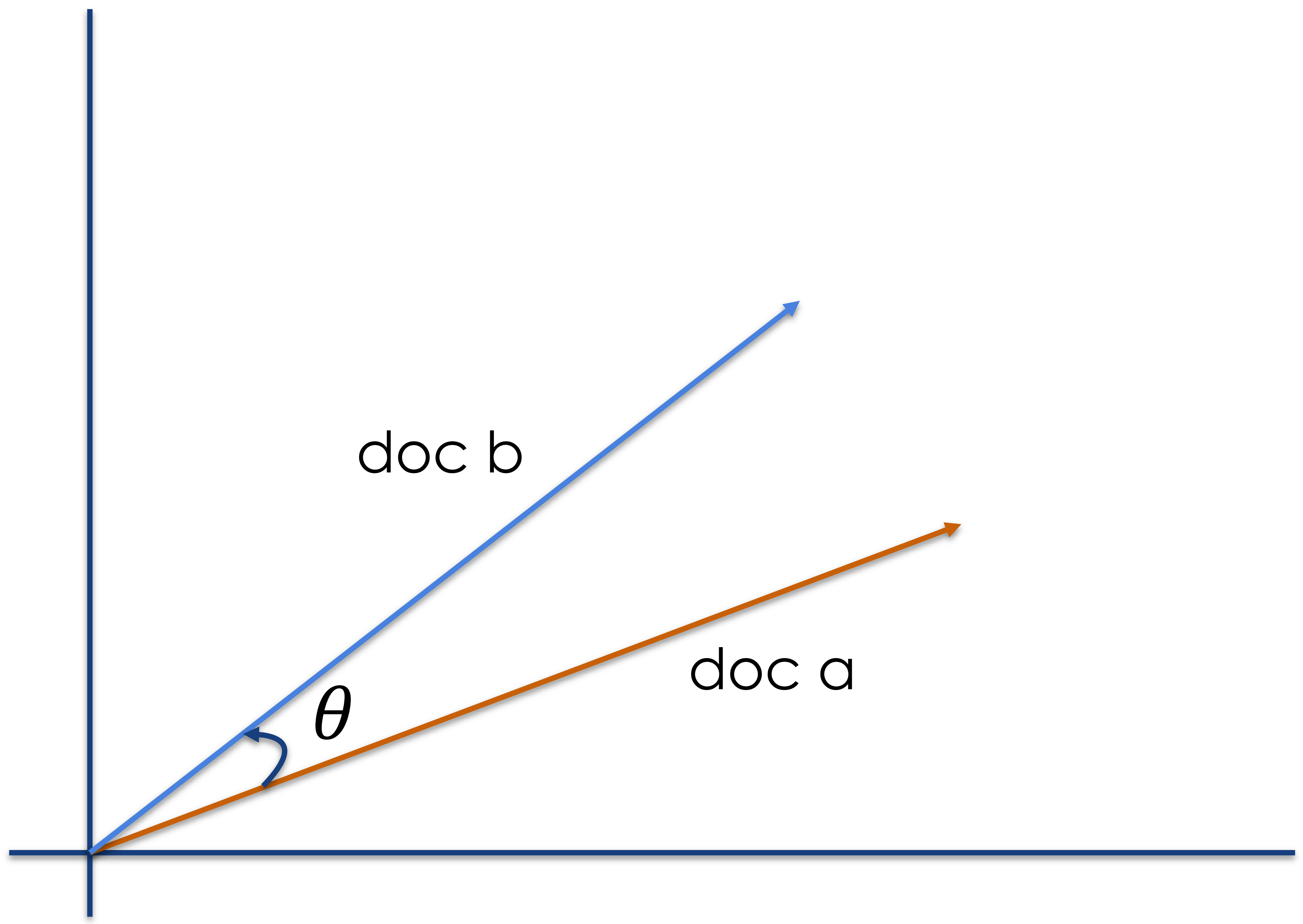
cosine similarity

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{\parallel \vec{a} \parallel \parallel \vec{b} \parallel}$$

length of feature vectors

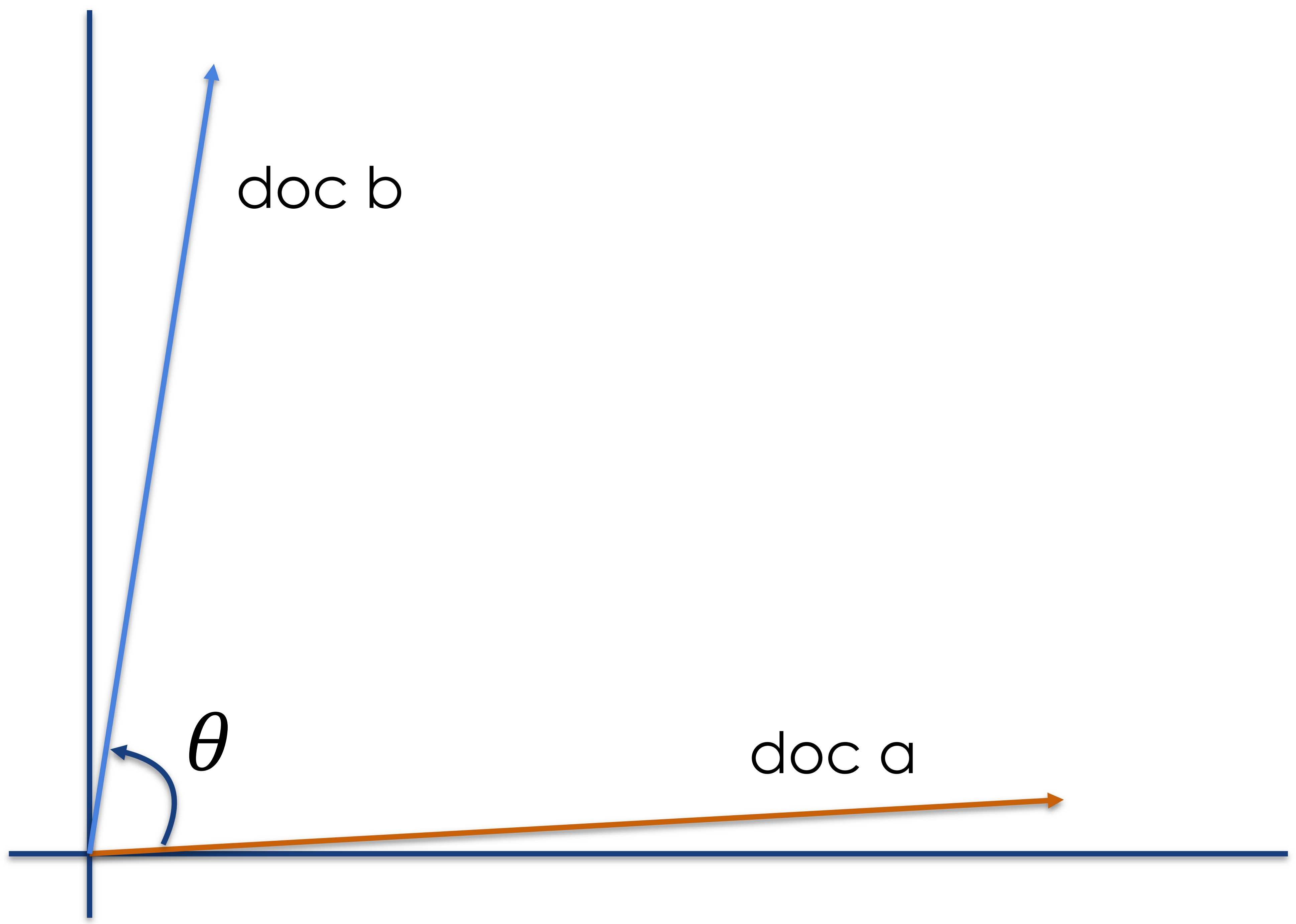
Cosine Similarity

- Two very similar documents yield a cosine similarity that is close to 1 (cos(0) = 1)



Cosine Similarity

- Likewise, two documents that are very distinct from each other yield a cosine similarity that is close to 0 (cos(90) = 0)



Cosine Similarity example

- Cosine Similarity can be for querying for relevant documents given a query string

- Let's say our query string is "cats and fish"

- First, we treat "cats and fish" as a document and perform TF-IDF on it

Cosine Similarity example

- Computed Normalized TF-IDF vector for "cat fish"

	big	cat	eat	fish	john
TF	0	1	0	1	0
Normalized TF	0	1/2	0	1/2	0
IDF	1.692	1.288	1.288	1.288	1.693
TF * IDF	0	0.644	0	0.644	0
Normalized TF-IDF	0	0.707	0	0.707	0

IDF of corpus

Cosine Similarity example

- Treating the Normalized TF-IDF vector for the query string "cats and fish" as a document within our corpus

	big	cat	eat	fish	john
doc1	0	0.606	0	0	0.796
doc2	0	0.817	0.409	0.409	0
doc3	0.681	0	0.518	0.518	0
query	0	0.707	0	0.707	0

Normalized TF-IDF

Cosine Similarity example

- How similar is the query string compared to doc2?
- Vector Multiplication of $\vec{Query} \cdot \vec{Doc2}$ = $\boxed{(0.707 * 0.817)}$ + (0.707 * 0.409) = 0.866
- Length Multiplication of $\| \vec{Query} \| \| \vec{Doc2} \|$ = 1 * 1 = 1
- Cosine Similarity = $cos\theta = \frac{\vec{Query} \cdot \vec{Doc2}}{\|\vec{Query}\| \|\vec{Doc2}\|}$ = 0.866 / 1 = 0.866
- Query and Doc2 have similar words, and that similarity is being reflected with a cosine similarity of 0.866, which is close to 1

	big	cat	eat	fish	john
query	0	0.707	0	0.707	0
doc2	0	0.817	0.409	0.409	0

Cosine Similarity (in code)

- Use sklearn for Cosine Similarity calculation
- Restructure code for reusability

```
import string
import pandas as pd
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

def preprocess(docs):
   cleansed = []
   punc = str.maketrans('', '', string.punctuation)

   for doc in docs:
      doc_no_punc = doc.translate(punc)
      words = doc_no_punc.lower().split()

      words = [lemmatizer.lemmatize(word, 'v')
                   for word in words if word not in stop_words]

      cleansed.append(' '.join(words))

   return cleansed
```

Cosine Similarity (in code)

- Fit the TF-IDF model using the terms found in our corpus
- Used the fitted model to generate feature vectors for the documents in the corpus and the query string

```
lemmatizer = WordNetLemmatizer()
stop_words = stopwords.words('english')

docs = [
    'John has some cats.',
    'Cats, being cats, eat fish.',
    'I ate a big fish'
]

query = ['cats and fish']

docs_clean = preprocess(docs)
query_clean = preprocess(query)

tfidf = TfidfVectorizer()
tfidf.fit(docs_clean)

fv_corpus = tfidf.transform(docs_clean).toarray()
fv_query = tfidf.transform(query_clean).toarray()
```

Cosine Similarity (in code)

- Viewing our query string's TF-IDF feature vector in a tabular format using Pandas

```
fv = pd.DataFrame(data=fv_query,
                  index=['query string'],
                  columns=tfidf.get_feature_names())

print(fv, '\n')
```

	big	cat	eat	fish	john
query	0	0.707	0	0.707	0

Cosine Similarity (in code)

- Performing a Cosine Similarity for the query string against all documents in our corpus

```
similarity = cosine_similarity(fv_query, fv_corpus)

cs = pd.DataFrame(data=similarity,
                  index=['cosine similarity'],
                  columns=['doc1', 'doc2', 'doc3'])

print(cs)
```

↓

	doc1	doc2	doc3
Cosine Similarity	0.428	0.866	0.366

Cosine Similarity (in code)

Corpus (stop-words removed & lemmatized)

- doc1: john cat
- doc2: cat cat eat fish
- doc3: eat big fish

Query String (stop-words removed & lemmatized)

- cat fish

- The Cosine Similarity values indicate that doc2 has the highest similarity with our query string, followed by doc1 and doc3

	doc1	doc2	doc3
Cosine Similarity	0.428	0.866	0.366

Application - Search Engine

- Rank documents by relevance given a query string
- Query String - “cats and fish”
- Returned Results
  - doc2 (rank: 0.866)
  - doc1 (rank: 0.428)
  - doc3 (rank: 0.366)

Entire code

```
import string
import pandas as pd
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

def preprocess(docs):
    cleansed = []
    punc = str.maketrans('', '', string.punctuation)

    for doc in docs:
        doc_no_punc = doc.translate(punc)
        words = doc_no_punc.lower().split()

        words = [lemmatizer.lemmatize(word, 'v')
                    for word in words if word not in stop_words]

        cleansed.append(' '.join(words))

    return cleansed


lemmatizer = WordNetLemmatizer()
stop_words = stopwords.words('english')
```

→

```
docs = [
    'John has some cats.',
    'Cats, being cats, eat fish.',
    'I ate a big fish'
]

query = ['cats and fish']

docs_clean = preprocess(docs)
query_clean = preprocess(query)

# compute normalized TF-IDF
tfidf = TfidfVectorizer()
tfidf.fit(docs_clean)

fv_corpus = tfidf.transform(docs_clean).toarray()
fv_query = tfidf.transform(query_clean).toarray()

fv = pd.DataFrame(data=fv_query,
            index=['query string'],
            columns=tfidf.get_feature_names())

print(fv, '\n')

#compute cosine similarity
similarity = cosine_similarity(fv_query, fv_corpus)

cs = pd.DataFrame(data=similarity,
            index=['cosine similarity'],
            columns=['doc1', 'doc2', 'doc3'])

print(cs)
```

Workshop 2

Cosine Similarity
(Rank Query-Results)

The End